Stop Heating an Empty Building: Reward Anatomy and a Supervisory Layer for the NeurIPS 2026 Smart Buildings Challenge

Semyon Morev
Institute of Computer Science
NUST MISIS, Moscow, Russia
m2508348@edu.misis.ru

Abstract

The Smart Buildings Challenge asks for one controller that runs, without retraining, in buildings it has never seen. Two facts about its reward decide the episode: comfort regret is the occupancy-weighted *mean* shortfall over present occupants, so readiness at the 06:00 mode change dominates; and because at least $30\,\%$ of the supply air is outdoor air while discharge is capped at 308.15 K, the gas billed per unit of delivered heat falls as the zone gets colder (2.0 at 294 K, 1.18 at 270 K). On Wu et al.'s public adaptive-physics controller we add a supervisory layer: CtrlW enforces a 2 h readiness constraint ($-1.8\,\%$ on 30 shipped scenarios), and CtrlW5 stops heating the empty building at night and delivers the recovery in one capacity-limited burst. With the 278 K ($4.9\,^{\circ}$C) night floor we ship—the coldest room measured stays at $+4.6\,^{\circ}$C—CtrlW5 scores 0.0071980 ($-2.3\,\%$ against CtrlW, $-2.6\,\%$ on 49 held-out buildings and climates) and 0.01032 on the leaderboard; without the floor it reaches -80.31 on the forum's seven-day anchor, past the best public number (-81.9). We then price the missing information—a zonal occupancy sensor $-2.9\,\%$, perfect per-zone foresight $-6.0\,\%$, a perfect 12 h forecast ≈ 0—and audit two open accounting seams, worth fractions of a unit to an honest controller but 37 anchor units to a boiler-billing exploit.

1 Introduction

The Smart Buildings Challenge [9] evaluates controllers under a protocol that makes transfer the whole problem: one policy is instantiated afresh for each of 30 scenarios—six buildings crossed with five weather stations—and must run 8,640 steps of five simulated minutes without retraining, network access or a site visit. Zone count varies threefold ($V \in [30, 85]$) and air-handler count threefold ($H \in [3, 9]$), so the action dimension, $3H + 2 + 2V$ continuous values plus $H + 1$ binaries, changes per scenario; no physics, plant sizes or per-zone occupancy are exposed; and submissions get 9 CPU-hours of wall-clock for all 30 scenarios [16], ten times the bang-bang baseline's wall-clock, about 125 ms per step. End-to-end reinforcement learning is therefore expensive—0.04–0.13 CPU-seconds per step on 1.2–2.6×10^5 control volumes means about 107 CPU-hours for 5×10^6 PPO steps, and the organizers report only single-digit-percent gains from SAC on one building [7]. Meanwhile the leaderboard is public, so the object to beat is a documented notebook: the strongest [24] reaches 0.01062 against 0.05498 for the organizers' `pid_water`. We port that controller bit-exactly and ask what is left to fix once a strong public controller is in hand.

The reward has a structure that decides the episode. Comfort is charged to *present* occupants only, and as a mean, so the first arrivals cost as much per kelvin as the peak: readiness at the mode change is a first-order objective and the night is free. The bill is envelope heat times a multiplier that

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the $30\,\%$ outside-air floor and the 308.15 K discharge cap fix (Eq. 2)—so the same recovered heat is cheaper when it is delivered into a colder building, in one burst, at the coil's cap. The consequence is a two-number change to a public controller. CtrlW adds a 2 h readiness constraint and reward-consistent fuel weights; CtrlW5 then stops heating the empty building, because the inherited 282 K standby floor, held by the coldest zone of each air-handler group, keeps every group's plant on from 19:00 to 04:00. Releasing it plus a cooler hot-water loop is worth 12.8 of the anchor's 93.1 units, and 6.4 with the 278 K ($4.9\,^{\circ}$C) floor we ship. A receding-horizon planner we built to look for the same gain does not find it, which is why we present this as a *supervisory sequencing* result rather than a control-quality one.

Contributions. **C1** a reward anatomy (comfort is a per-occupant mean; the bill is envelope heat times a closed-form ventilation multiplier; the exchange rate between them is $\approx 39{:}1$ against trading the occupied band); **C2** CtrlW, a supervisory layer on the public controller, with the diagnosis that its preheat horizon was silently capped at one hour; **C3** CtrlW5, the night release—its decomposition, a search over the same 16-knob family and a floor sweep that agree on it, and the frost-floor cost curve that prices the limit we ship; **C4** an evaluation on 30 shipped scenarios, 49 holdouts, the corrected tree, reseeded occupancy and the leaderboard; **C5** a measured value of information; **C6** an audit of the two seams plus the exact fused simulator step ($2.1\times$) these studies ran on.

2 The benchmark, and related work

A scenario is a partially observed Markov decision process over a temperature field, plant parameters and hidden per-zone occupancy; the policy acts in $\mathcal{A} = \mathbb{R}^{3H+2+2V} \times \{0,1\}^{H+1}$ (per handler a supply-air temperature, a static-pressure setpoint and a recirculation fraction $r_k \le 0.7$; two plant values; per zone a damper floored at $10\,\%$ and a reheat valve), observes $6V + 15H + 34$ values with zone readings lagged one step and *one* building-level occupant count, but no per-zone occupancy and no weather forecast; the plant cools centrally and heats only through per-zone reheat coils. With live setpoints $T_v^{\mathrm{lo}}, T_v^{\mathrm{hi}}$ and per-zone shortfall $s_{v,t}$ (linear to $w = 5$ K, quadratic beyond), the step reward and score are $r_t = -0.6\,C_t - 0.2\,\hat{E}_t - 0.2\,\hat{C}_t \le 0$ and $S = \frac{1}{30}\sum_{s=1}^{30} \left|\frac{1}{T}\sum_{t=1}^{T} r_{s,t}\right|$, with C_t the occupancy-weighted shortfall of Eq. (1) and $\hat{E}_t, \hat{C}_t$ metered electricity-plus-gas cost and carbon normalized by the draw at rated power times H; lower S is better. Regret is linear to 5 K then quadratic, so one runaway zone dominates an episode, and the denominators scale with H, so one parameter set behaves consistently across building sizes.

The challenge [9] asks whether domain-specific models or generic foundation models adapt better to unseen buildings, on the substrate of the Smart Buildings Control Suite [7] with its lightweight finite-difference simulator [10]; its authors concluded that aggregating a policy into a static schedule retains most of its value [6] and that better simple heuristics remain to be found [8], which is the line our result continues. Because the leaderboard is public we measure against three notebooks rather than re-derive them—a per-zone PI rule [13], a residual PPO policy on it [4], and the analytic one-step search we adopt [24]—all rule-based or one step deep. Transfer is formalized by Building2Building [22] and attacked by GenTL [18] and BUILDA [12], whose generator we reuse; MPC's bottleneck is modelling effort rather than optimization [5], which is why we identify per zone online and optimize nothing per step, with Guideline 36 [2] and trim-and-respond [23] supplying the reset semantics of the base controller. Evolution strategies suit a slow deterministic simulator because rollouts parallelize exactly [11, 19]; CityLearn 2022 [17] is the closest precedent, and benchmark hygiene follows BOPTEST [3] and [15].

3 The reward has a structure

Comfort regret is a per-occupant mean, so arrival dominates. For zone v with live band $[T_v^{\mathrm{lo}}, T_v^{\mathrm{hi}}]$ and shortfall $s_{v,t}$ (linear to 5 K, quadratic beyond), the productivity term of the reward is

$$\frac{P - P^{\min}}{P^{\max} - P^{\min}} - 1 \;=\; -\,\frac{\sum_v o_{v,t}\, s_{v,t}}{\sum_v o_{v,t}}, \tag{1}$$

the occupancy-weighted *mean* shortfall over the occupants present [9]. The headcount cancels: a zone out of band with two people costs as much per kelvin as the same zone failing with sixty.

Occupancy is exogenous—two occupants per zone, arrivals 06:00–12:00, departures 13:00–19:00—so the ramp runs from 0 to $2V$ people across the morning (74 on the anchor) and a reactive controller pays a fixed price at every mode change. The measurement is one horizon change: the public base controller's preheat gain is multiplied by a parameter that lives inside a 60 min look-ahead flag, so its ramps never exceeded one hour. Moving the ramp outside the flag at a 2 h horizon cuts mean hour-06 comfort regret from 46.2×10^{-4} to 9.2×10^{-4} per step and the score from 0.0075040 to 0.0074263; the remaining $0.0074263 \rightarrow 0.0073694$ comes from applying the reward-consistent weights on top of it, an interaction the one-at-a-time ablation does not capture. Longer horizons keep improving the arrival hour while the score rises (0.0074906 at 3 h, 0.0075614 at 4 h): readiness saturates once every zone is in band; the gas does not.

The bill is envelope heat times a ventilation multiplier. On the public forum's comparison episode (`building_train_1/fairbanks_1`, 2 016 steps, mean outside $-26.3\,^{\circ}$C) the same controller loses 93.12 units: comfort 0.54 (0.6 %), energy 26.23 (28.2 %), carbon 66.35 (71.3 %), carbon being 2.5 times energy. Instrumenting the trajectory at the model call partitions the bill: 73.3 % envelope reheat at the band bottom, 22.9 % the outside-air share of the same reheat, 1.9 % fans, 1.3 % boiler standing loss, 0.6 % pump, no chiller. That outside-air share is not a modelling choice but the action space's floor: recirculation is capped at $r \leq 0.7$, so at least 30 % of the supply air is outdoor air, and the coil must lift it from $T_{\text{mix}} = 0.7\,T_{\text{ret}} + 0.3\,T_{\text{out}}$ to the discharge cap $T_{\text{dis}} = 308.15\,\text{K}$ while the zone only receives the drop from T_{dis} to T_{zone}. With $T_{\text{ret}} \approx T_{\text{zone}}$ the plant's own air-side law is

$$\frac{\text{gas billed}}{\text{net zone heat}} \;=\; 1 + \frac{0.3\,\big(T_{\text{zone}} - T_{\text{out}}\big)}{T_{\text{dis}} - T_{\text{zone}}}, \tag{2}$$

so at the anchor's ambient the multiplier is 2.00 for a zone held at 294 K and 1.18 for one at 270 K (Figure 1a). *Heat delivered into a colder building is cheaper per delivered watt.* What a controller cannot change is the 0.3; what it can is the temperature at which it delivers the recovery. Measured over the episode, gas per net zone watt is 1.975 for CtrlW and 2.061 for CtrlW5 against a cap-limited 1.949 and 2.045: the air side has about one unit of slack in total (1.15 units, 1.2 %), so *airflow is not where the money is*. The ventilation share is large in the counterfactual sense (coil heat against non-ventilation load, a factor 2.24) but that is a partition of the bill, not a lever.

What the anatomy rules out. The exchange rate between the two terms forbids trading the occupied band: on the anchor 624 of the 2,016 steps have at least one occupant, and a uniform 1 K shortfall there costs $0.6 \times (1/5) \times 624 \approx 75$ reward units (the comfort term is a mean over present occupants, so it is counted per occupied step, not per occupied zone-hour), while one kelvin of building temperature is 0.29 MWh of net heat = 0.57 MWh of gas ≈ 1.9 units—an ≈ 39:1 ratio. Measured: holding the schedule's own setpoint instead of 0.12 K above it costs 1.53 units, and hours 04–08 carry 41.5 % of the anchor's energy-plus-carbon loss in 21 % of the hours. What is left is the band nobody occupies: the reward charges comfort only to present occupants, so the night is free—and CtrlW, like the family it comes from, heats it anyway.

4 Method: from a public controller to a night release

4.1 The base controller

We adopt the public controller of Wu [24], credit it as such, and reproduce it exactly: with all our flags off, our port returns its dev30 score to the last digit (0.0075040). It identifies one first-order zone model by recursive least squares with a forgetting factor, lagged one step to match the sensor delay; evaluates candidate (supply-air temperature, recirculation) pairs analytically against those models per handler, a one-step-deep MPC costing $O(K)$ closed-form evaluations; corrects drift with a small MLP ensemble; and weights predicted gas 0.30 against electricity. It is strong—comfort 2.14×10^{-4} per step on dev30—but it plans neither the morning nor the night: its standby policy holds each zone between 282 and 307 K, and because a group's plant runs when *any* zone of the group calls, one cold zone holds every group's plant on all night.

4.2 CtrlW: readiness, weights, guards

Readiness constraint. The base controller ramps its preheat setpoint by a gain multiplied by `precondition_hours`, but the multiplication lives inside the code path guarded by `comfort_mode_soon`—a flag true only in the 60 minutes before comfort-mode onset—so any horizon longer than 1 h was clamped by construction. We implement the ramp outside the flag, as a supervisory decision taken once per simulated hour from the identified models: at the mode change t_s, every zone that may be occupied must satisfy

$$T_v^{\text{lo}} + \mu \;\le\; \hat{T}_v(t_s) \;\le\; T_v^{\text{hi}} - \mu, \qquad \hat{T}_v(t_s) = a_v^{t_s - t} T_{v,t} + \big(1 - a_v^{t_s - t}\big) T_v^{\text{eq}} + b_v^{\top} \bar{x}, \tag{3}$$

with μ a lag margin, T_v^{eq} the identified equilibrium under the preheat command and $\bar{x}$ the last exogenous regressors. The controller preheats from $t_s - h$ and holds the plant enabled until the constraint is met; $h = 2$ h is the tuned horizon.

Weights and guards. Per unit of gas energy the reward charges $0.2\,(p_g + \lambda\kappa_g)$ and per unit of electricity $0.2\,(p_e + \lambda\kappa_e)$ (p price, κ carbon intensity, λ the ratio of the cost to the carbon normalizer, up to a common scale), so the consistent relative weight is $(p_g + \lambda\kappa_g)/(p_e + \lambda\kappa_e) \in [0.443, 0.509]$ on the 30 shipped scenarios, mean 0.481: the base under-weights gas by about $40\,\%$. Measured, it is a plateau (0.0075035, better in 16 of 30) because gas is spent on reheat during warm-up and electricity on fans and chillers during the day; we adopt the derived weight because it is the correct objective and claim no gain for it. Two guards are inert: an occupancy-aware release never fires on dev30 (occupancy never returns to zero inside the comfort window) and a holiday guard never fires (no holiday in the shipped scenarios); both leave the score at 0.0075040 exactly, and we say so rather than credit them.

4.3 CtrlW2 and CtrlW4: what tuning the class can buy

CtrlW2 replaces the fixed horizon with a per-zone one, $\text{clip}(\text{safety} \cdot \ell_v + \text{slack}, 1, 3)$ h, where ℓ_v is the lead the zone's own model needs under full conditioning: 0.0073576 ($-0.16\,\%$, better on 12, worse on 6), inside the reseed spread. *CtrlW4* adds the legal half of our oracle study—a fixed 2 K predawn lift for zones on the standby floor, gated on the clock and on a comfort day coming—scoring 0.0073445 ($-0.34\,\%$, better on 8, worse on 10) and 0.0071587 corrected, and inert on the anchor (-93.13 against -93.12). Both are small; the next subsection is not.

4.4 CtrlW5: do not heat an empty building

The night is unoccupied by construction (arrivals 06:00–12:00), so comfort charges nothing there, yet CtrlW holds each group's coldest zone at 282 K from 19:00 to 04:00. CtrlW5 changes exactly two numbers in `PARAMS`: `standby_minimum` $282 \to 262$ K, which releases the floor, and `hot_water_target` $345 \to 336.32$ K. Restoring them reproduces CtrlW4 bit-exactly, and the two are worth 12.07 and 0.75 units of the 12.82: the single floor release is the mechanism, the cooler loop is a bonus; the module needs numpy only, no building or weather identifier, no forecast and no occupancy channel. Three mechanisms, all measured on the anchor (Table 2, Figure 1): *(a) the night heating stops*—night coil duty falls from 30 kW to 0 kW while the building's *mean* temperature barely moves ($287.9 \to 287.4$ K, i.e. $14\,^{\circ}$C, still 2 K from the schedule's own 289 K night setpoint: it is the coldest zones and the group plant that change, not the building); *(b) the recovery is one burst at the discharge cap*—by Eq. (2) heat delivered into a colder mass is cheaper per delivered watt, and CtrlW5's discharge sits at $35.00\,^{\circ}$C with 1.1 K mean headroom against CtrlW's 2.3 K, its gas per net zone watt 2.061 against a cap-limited 2.045 ($0.7\,\%$ of air-side slack, against CtrlW's $1.2\,\%$); *(c) the stored heat pays for the day*—charging the mass at 04:00–06:00, the cheapest hours (88 kg CO_2/MWh), and letting it drift through the band costs nothing in comfort, because the band's upper edge is free, and removes much of the continuous daytime coil duty.

Net: gas $27.19 \to 23.27$ MWh ($-14.4\,\%$), coil heat $26.72 \to 22.97$ ($-14.0\,\%$), ventilation-attributable load $14.79 \to 12.50$ ($-15.5\,\%$), comfort $-0.54 \to -1.12$. The $+12.82$ units decompose into envelope reheat $+8.75$, the outside-air share $+4.42$, fans $+0.27$, pump -0.22, boiler standing loss $+0.18$, comfort -0.58; by hour block, $+3.3$ (00–04), $+2.7$ (04–06), $+0.7$ (06–12), $+3.1$ (12–19), $+3.0$ (19–24) (Figure 1b). A search over the same family and a floor sweep agree that this is the family's optimum rather than a lucky knob, though neither rules out a better mechanism

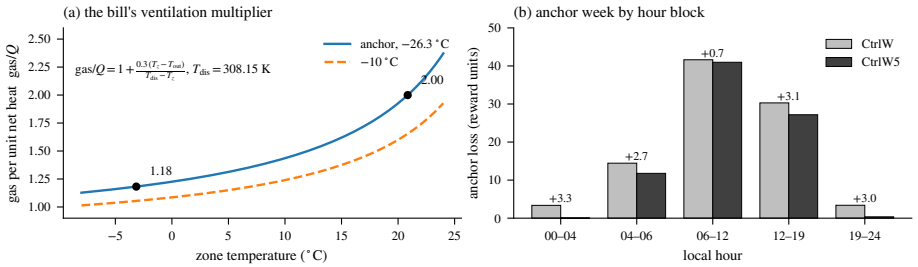


Figure 1: **(a)** The bill's ventilation multiplier, Eq. (2), at the anchor's ambient ($-26.3\,^{\circ}$C) and at $-10\,^{\circ}$C: the same delivered heat costs 2.00 units of gas at a 294 K zone and 1.18 at 270 K, which is why releasing the night and recovering in one burst pays. **(b)** The anchor's loss by hour block: releasing the floor removes 3.3 (00–04), 2.7 (04–06) and 3.0 (19–24) units from the blocks the plant used to heat, and 0.7 and 3.1 from the two daytime blocks, where the stored charge replaces coil duty. Bar heights are losses (lower is better) and the labels are savings.

outside it. A 150-evaluation CMA-ES over CtrlW4's 16 supervisory knobs reaches -83.109 with its optimum on the lower bound of `standby_minimum`, and releasing that bound takes it to -80.298, i.e. to CtrlW5-262's -80.306 within 0.008 units. And the floor sweep is monotone and saturates at 262 K, where the floor stops binding and the building free-floats: the optimum is not "a deep setback", it is *no overnight heating at all*, with the comfort term flat at -1.35 across the sweep, so none of it is bought from the occupied band.

4.5 Frost protection, and what it costs

A released building in Fairbanks does not stay above freezing on its own: the 262 K release gains 12.8 units by letting the coldest room reach $-8.58\,^{\circ}$C on the anchor and $-11.2\,^{\circ}$C on a 30-day episode, with 19 and 90 hours in which some room is below $0\,^{\circ}$C. The simulator has no freezing model and the score has no penalty for it, so this is a choice, not a constraint. We ship a floor: **278 K**, i.e. a $4.85\,^{\circ}$C setpoint floor, which keeps the coldest room at $+4.66\,^{\circ}$C on the anchor and $+4.56\,^{\circ}$C on the three cold 30-day scenarios, with zero room-hours below $0\,^{\circ}$C, and still beats CtrlW by 6.9 % on the anchor and 3.8 % on the six Fairbanks scenarios (Appendix G). This is a measured minimum on four scenarios, not a guarantee: the floor is a setpoint the controller can miss. The price is steep at the top—282→280 K buys 3.5 units, 280→278 another 2.9—and flat at the bottom (Figure 3); $\geq 5\,^{\circ}$C needs 280 K and costs 9.3 of the 12.8 units, and 274 K leaves rooms at $+0.63\,^{\circ}$C, inside the pipe-freezing margin the limit exists to protect. Holding the floor is also the cheaper way to keep it: a per-zone latch that lets a room reach 278 K and recovers it in one burst to $278+b$ K loses 0.32–4.32 units for $b = 0$–4 K and lets the room dip to $+1.1$–$3.3\,^{\circ}$C.

5 Results

Protocol. `dev30` is the 30 shipped scenarios at full episode length (8,641 five-minute steps), which is what the leaderboard draws from; `gen` is 32 scenarios from eight generated buildings under weather windows absent from tuning; `ext` replays those buildings and two shipped ones under 17 Meteostat 2023 stations absent from the benchmark (Appendix I). Scores are means over scenarios of $|\frac{1}{T}\sum_t r_t|$ and comparisons against a base are paired. Episodes are 8,640 scored steps plus a terminal one (result files report 8,641). Reseeding occupancy moves a score by at most 0.18 % and a paired gain by at most 0.12 points (§5, below), so we read only larger deltas as gains. Numbers are shipped unless marked corrected; we report both, because final verification runs the corrected simulator [20, 21] and reseeds occupancy, and neither seam nor seed is something we tune against.

Main comparison. Table 1. On dev30 the organizer baseline scores 0.05498, the per-zone PI rule 0.01042, the base controller 0.0075040, CtrlW 0.0073694 and CtrlW5-278 0.0071980; the ordering is unchanged on the corrected tree, which moves the six non-organizer controllers by -1.8 to -3.2 % and makes `pid_water` worse. The kit's other reference controllers are weaker still by its own shipped returns—fixed setpoint 0.0659, PI 0.0674, bang-bang 0.2252—a file that puts `pid_water` at 0.05414 against our re-run's 0.05498, so read them as indicative. On the holdouts CtrlW5-278

Table 1: Main comparison; score S (lower is better). `dev30` is 30 shipped scenarios at 8,641 steps, `gen` 32 held-out generated scenarios and `ext` 17 unseen climates; *shipped/corrected* are the two simulator trees. CtrlW5-278 is the controller we ship (278 K night floor) and CtrlW5-262 the unconstrained bound. Public rows are the authors' leaderboard scores unless we re-ran the controller; shipped `pid_water` combines 25 scenarios at 0.054363 with 5 re-runs at 0.058058.

Method	dev30 shipped	dev30 corrected	gen / ext	Kaggle public
`pid_water`	0.05498	0.05789	0.03620 / 0.05189	—
per-zone PI [13]	0.01042	0.01023	0.007673 / 0.010748	0.01420
Wu et al. [24]	0.0075040	0.0072641	0.005293 / 0.008015	0.01062
CtrlW	0.0073694	0.0071449	0.005179 / 0.007836	0.01051
CtrlW4	0.0073445	0.0071587	0.005170 / 0.007813	0.01049
CtrlW5-278	**0.0071980**	**0.0070409**	0.005058 / 0.007616	0.01032
CtrlW5-262	0.0071338	0.0069430	0.005052 / 0.007577	0.01024

is $2.6\,\%$ better than CtrlW—better on 38 of 49 scenarios, identical on 11, worse on none (sign test $p < 10^{-11}$)—so the gain is not a dev30 artefact; the unseen climates are $51\,\%$ worse than the same buildings under in-distribution weather (0.007616 against 0.005058), so the transfer gap is climatic, not architectural. On the public leaderboard CtrlW scored 0.01051 (submission 58025908), CtrlW4 0.01049, CtrlW5-278 0.01032 (58059955) and CtrlW5-262 0.01024 (58050174) against the base controller's 0.01062; the leaderboard/offline ratio is 1.36–1.44 across the six methods with both numbers (per-zone PI 1.36, Wu 1.42, CtrlW 1.43, CtrlW4 1.43, CtrlW5-278 1.43, CtrlW5-262 1.44), a consistency check rather than a calibration. The frost floor costs $0.8\,\%$ on the leaderboard (0.01032 against 0.01024), as it costs $0.9\,\%$ on dev30.

The anchor. On the seven-day episode the public forum compares on (2 016 steps, mean outside $-26.3\,^{\circ}$C, Appendix A) CtrlW scores -93.12, CtrlW4 -93.13, and CtrlW is 11.2 units behind the best public physically consistent controller (-81.9; a twin-plus-LP planner gets -85.4, a free-action LP bound -79.1). Neither information ($+2.60$ units for perfect per-zone foresight) nor the accounting seams (§7, fractions of a unit) closes that; and our own 8 h receding-horizon LP on a 79-node twin moves $7.6\,\%$ the wrong way (-100.20), because it *buys* night capacity margin instead of selling it (night bucket $-1.22 \to -9.33$, day plan $+2.41$). Fixing its three defects—heat priced at its physical cost, the coil's cold side at the zone rather than the mixed-air temperature, a plan margin and a night ceiling—recovers $+6.05$ of the 7.08 and leaves it 1.0 unit behind CtrlW (-94.15); when the twin is finally allowed to float the night it still does not discover the release (-98.53). The 12.8 units are a *sequencing* decision: released, the same controller reaches -80.31, and -86.68 with the frost limit we ship—our own 8 h LP, with a better model and three defects fixed, did not find it.

Ablations, sweeps, reseeds, and what loses. Table 3 (appendix) adds one change at a time: the readiness horizon is the only large one at that stage, the two guards are inert and the fuel weight a plateau. The horizon sweep is non-monotone in the score—hour-06 comfort falls monotonically while the score does not (0.0075040, 0.0074263, 0.0074906, 0.0075614)—because readiness saturates once every zone is in band and each extra hour buys gas at night instead of comfort at dawn. Reseeding occupancy moves CtrlW's dev30 score from 0.0073694 (seed 17321) to 0.0073726 (12345) and 0.0073743 (777), and CtrlW5-278's from 0.0071980 to 0.0072012 and 0.0072111 ($+0.18\,\%$ at most), while CtrlW5-278's gain over CtrlW reads -2.33, -2.33 and $-2.21\,\%$ under the three seeds. Three things lose: holding the schedule's own setpoint instead of 0.12 K above it costs 1.53 units on the anchor and $4.3\,\%$ on five of six Fairbanks scenarios, every one worse; a deeper night setback in the *same* direction as the base controller's (285 K) costs 6.9 units and 278 K through the band-clamp path 96.8—the two levers tried before the release; and the receding-horizon LP, as above.

6 What information is worth

The observation space exposes one occupant count for the whole building, no per-zone occupancy and no weather forecast, while comfort is charged per zone. We hand the same controller each missing channel in turn—a per-zone occupancy sensor, a perfect weather forecast, the realised schedule, Table 4—and never submit any of them, so the deltas bound the value of sensing rather than propose an entry. An oracle is an instrument only if it is transparent, and these are: a policy that receives the

payload and drops it reproduces an ordinary run to the last digit on two full episodes, so the channel perturbs neither the occupancy draw nor the clock, and the forecast is exact rather than noisy (it equals the sensor reading h hours later to 1.4×10^{-5} K, the float32 quantisation of the observation).

Zonal occupancy pays $2.9\,\%$. Dropping each zone to standby 15 min after its own occupants leave, rather than waiting for the whole building to empty as CtrlW's occupancy release does, improves full dev30 from 0.0073694 to 0.0071585 and the corrected tree from 0.0071449 to 0.0069403 (both $-2.86\,\%$); on the ten-scenario subset it is $-2.30\,\%$, better on 10 of 10, recovering 23.9 reward units of energy plus carbon for 0.16 units of extra comfort regret. Raising the release floor from 282 to 289 K moves the subset score by $0.09\,\%$ and keeps 10/10, so the result is about *which* zones are empty, not how cold they get.

A perfect forecast is worth nothing on top of a clock rule. A fixed 2 K pre-dawn lift, gated only on the clock, the comfort-day predicate and the zone sitting on its standby floor, scores 0.0073463 on dev30 ($-0.31\,\%$), and 0.0071582 against CtrlW's 0.0071449 on the corrected tree—a wash there (the same lift shipped inside CtrlW4 scores 0.0073445, the difference being the adaptive horizon underneath). Driving that lift and the preheat horizon from a perfect 12 h forecast scores 0.0073572: $0.15\,\%$ *worse* than the constant lift on dev30 and $0.01\,\%$ better on the subset, so the information is worth zero to within our resolution and can have either sign. The mechanism is gated on the zone's own state and on a comfort day coming, and the horizon response is flat-topped around 2 h; on both cooling stations the forecast variant is bit-identical to the no-information one.

The schedule is worth twice the sensor, and the difference is the lag. Per-zone foreknowledge of the episode's own occupancy, used to give each zone its own window with a 5 h preheat lead, scores 0.0112001 on the subset ($-6.04\,\%$, 10/10) and -90.52 on the anchor ($+2.60$ units, $+2.79\,\%$), with $+0.48$ to $+2.76\,\%$ at leads of 2–8 h; the same information delivered five minutes late and without a calendar loses 3.7 percentage points. Relaxing the band by 1 K in zones nobody has entered yet is the asymmetric bet against it: $-0.44\,\%$ of score, but only 5 of 10 scenarios and comfort regret multiplied from 5.33 to 39.64.

7 Benchmark integrity

The reward is computed from a simulator whose cost accounting is not closed under the policy's action space. Two seams are public and both have a proposed fix: the hot-water tank is billed from its setpoint rather than from its burner [20], and the return air is mixed by floor area rather than by airflow [21]. Neither fix is merged and the leaderboard runs the unfixed tree, so what each seam is worth—to whom, and whether it survives correction—is a measurement, taken at the model calls by an instrument that leaves every per-step reward bit-identical and reproduces the published return-air fix exactly (Appendix D).

B1 is a gift an honest controller does not take. A probe applying the published setpoint recipe on top of CtrlW delivers the same $26.72\,\mathrm{MWh_{th}}$ of reheat while 11.1 MWh of gas stops being billed, comfort does not move (-0.539 against -0.541), and the anchor improves by $+36.93$ units ($+39.7\,\%$)—3.3 times the gap to the best public score on the same episode. It is pure accounting: on the corrected tree the same probe is 0.59 units *worse* than doing nothing special, because a tank that actually cools delivers colder water. Across dev30, gated to the base controller's own heating periods, the recipe takes CtrlW5-278 from 0.0071980 to 0.0058864 ($-18.2\,\%$; $-36\,\%$ on Fairbanks, 0 in Singapore) and costs $0.46\,\%$ corrected: a fifth of an honest score is free to whoever cycles the tank. An independent residual-PPO entry found the same seam without being told of it: over 30 Fairbanks days its policy drew 914 kWh more tank heat than it was billed for, $39\,\%$ of its gain [4].

B2 is worth nothing to a controller that moves air well. The manufactured heat is 0.7–$1.0\,\%$ of the coil heat the three well-scoring controllers buy, and the artifact is worth *minus* a fraction of a unit (-0.34, -0.42, -0.86; Table 5). Harvesting it is priced out by the fan and by comfort: the trade clears only above $\Delta T \approx P/282$ K against the 0.12–0.59 K the field supplies. All three controllers improve when the accounting is fixed, and more so on a mild scenario ($10.4\,\%$ on a full-length Atlanta episode), while the probe reverses there: B1's worth is scenario-dependent and the anchor figure is a ceiling.

The third public channel is a leak rather than a seam: the occupancy generator uses one seed everywhere and the seed is not exposed, so replaying it yields the per-zone schedule, worth the foresight number of §6 ($2.79\,\%$). Final verification reseeds, so we never use it.

The anchor gap was a control question, and the answer is sequencing. On the seven-day anchor CtrlW scores -93.12 against -81.9 for a physically consistent public controller [20], -85.4 for a twin-plus-LP planner and -79.1 for a free-action LP bound on the linearised building [1] (a bound, not a controller). Neither information (the strongest channel, perfect per-zone foresight, is worth 2.60 units against the 11.22-unit gap to -81.9, so 4.3 times too little), nor B2 (a fraction of a unit), nor B1 (-0.3 units for the PR's own controller, $-81.9 \to -81.6$, as its author reports) explains it, and an 8 h receding-horizon planner moves $7.6\,\%$ the wrong way. What explains it is §4.4: the family heats the empty building at night. Releasing the floor takes the same controller to -80.31—1.6 units better than the best public number we know of, though worse than the free-action bound, which is an optimum and not an entry—with the frost-limited version we ship at -86.68. We therefore report the gap as *closed by scheduling for the controllers we can compute this on*, and attribute nothing to other teams beyond their published numbers: the audit bounds what these seams can be worth to a controller that respects comfort, not what any particular team did.

8 Tools: an exact fast simulator

Every claim above compares controllers on the same trajectory, so these studies are only as good as the number of episodes they can afford: one step costs 28.1 ms of CPU on the smallest building (thermal solve 13.7, rest of the HVAC step 6.0, reward and observation assembly 8.4, plus the policy's 1.8 outside `env.step`), against the ≈ 125 ms per step the 9-hour rule allows. Both halves must fall, and neither may change the trajectory: a speed-up that flips a damper decision is a different experiment.

The whole of `env.step`—zone means, thermostat decisions, the damper floor, supply-air temperature, boiler and hot-water demand, gains, the finite-difference solve with its Woodbury air side, the air-side fold-back, convection, the observation vector and the reward numbers—now runs as one C call, with the policy-facing half (≈ 0.4 ms) and the array API (≈ 0.2 ms) left in Python. Two simulator behaviours dictated the design (the boiler's sensor mutates the tank on every read, and zone thermal power is assigned rather than accumulated), and the kernel reproduces both. The gate is twelve closed-loop episodes (three buildings, both trees, two occupancy seeds) with observations and rewards bit-identical in all twelve ($\max|\Delta r_t| = 0.0$; zone temperatures $\leq 1.2 \times 10^{-15}$ relative, i.e. the policy-visible quantities are bit-identical, not the field), plus full 8 641-step traces on both trees identical to the last bit. Cost per step falls from 20.5 to 9.6 ms on the 30-zone building ($2.1\times$), $49.8 \to 27.8$ ($1.8\times$) on the 79-zone one and $67.6 \to 42.0$ ($1.6\times$) on the 85-zone one; five pure-Python patches recover 1.6–$1.9\times$ where C is unavailable. The shipped SuperLU is already optimal for a single right-hand side, batching is worth 4.1–$4.9\times$, a correct GPU solver loses to one CPU core, and a 200-mode reduced model drifts 1.0–1.2 K yet reproduces a five-way policy ranking exactly—a shortlisting filter, not a simulator (Table 6, Appendix E).

9 Limitations and conclusion

The release relies on an empty building: it pays because the challenge's occupancy model puts no arrivals between 19:00 and 06:00, and with night occupancy—or a hidden scenario colder than the coldest we ran—the unconstrained version is unbounded risk for at most 6 units, which is why we ship a 278 K floor and report its price. It is also cold-climate: inert on the warm stations, whose air side already sits at its action floors (Appendix F). *The anchor overstates the effect*: the same two numbers gain $13.8\,\%$ over seven days but $2.4\,\%$ over the 30-day episode of the same building, and 30-day Fairbanks deltas run $+2.2$ to $+11.8\,\%$. *The ceiling is the base model*: three of the four gains we report are under $2\,\%$, and the large one is large because it stops doing something.

Once a strong public controller is in hand, what is left to fix is not the control law but the *schedule of the plant*: stop heating at night and deliver the recovery in one burst at the coil's cap—12.8 of the anchor's 93.1 units, 6.4 with the 278 K floor we ship, and $0.0073694 \to 0.0071980$ on dev30.

References

[1] Houssem Alayet. Twin-based LP planner and free-action bound on the seven-day anchor. Kaggle discussion post 752142, NeurIPS 2026 Smart Buildings Challenge, 2026. `https://www.kaggle.com/competitions/neurips-2026-smart-buildings/discussion/752142`.

[2] ASHRAE. *ASHRAE Guideline 36-2024: High-Performance Sequences of Operation for HVAC Systems*. American Society of Heating, Refrigerating and Air-Conditioning Engineers, Atlanta, GA, 2024.

[3] David Blum, Filip Jorissen, Sen Huang, Yan Chen, Javier Arroyo, Kyle Benne, Yanfei Li, Valentin Gavan, Lisa Rivalin, Lieve Helsen, Michael Wetter, et al. Building optimization testing framework (boptest) for simulation-based benchmarking of control strategies in buildings. *Journal of Building Performance Simulation*, 2021. doi: 10.1080/19401493.2021.1986574.

[4] Alex Chilton. Yes, RL can run a building: Residual PPO. Kaggle notebook `alexchilton/yes-rl-can-run-a-building-residual-ppo`, NeurIPS 2026 Smart Buildings Challenge, 2026. Public leaderboard 0.01053; `https://www.kaggle.com/code/alexchilton/yes-rl-can-run-a-building-residual-ppo`.

[5] Ján Drgoňa, Javier Arroyo, Iago Cupeiro Figueroa, David Blum, Krzysztof Arendt, Donghun Kim, Eduard P. Ollé, Juraj Oravec, Michael Wetter, Draguna L. Vrabie, and Lieve Helsen. All you need to know about model predictive control for buildings. *Annual Reviews in Control*, 50: 190–232, 2020.

[6] Judah Goldfeder and John Sipple. Reducing carbon emissions at scale: Interpretable and efficient to implement reinforcement learning via policy extraction. In *Proceedings of the 11th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation (BuildSys '24)*, pages 403–407, 2024. doi: 10.1145/3671127.3699535.

[7] Judah Goldfeder, Victoria Dean, Zixin Jiang, Xuezheng Wang, Bing Dong, Hod Lipson, and John Sipple. The smart buildings control suite: A diverse open source benchmark to evaluate and scale hvac control policies for sustainability. *arXiv preprint arXiv:2410.03756*, 2025.

[8] Judah Goldfeder, Gabriel Guerra Trigo, Philippe Martin Wyder, Neil Kachappilly, and Hod Lipson. Evaluating performance of reinforcement learning agents to control buildings efficiently. In *2025 IEEE/CVF International Conference on Computer Vision Workshops (ICCVW)*, pages 2903–2909, 2025.

[9] Judah Goldfeder, Ayush Rai, Philippe M. Wyder, Na Li, Victoria Dean, Donna Vakalis, Jiong Lin, Bing Dong, Michael Rossetti, Nathan Kutz, Sohei Arisaka, John Sipple, Hod Lipson, and Yoshua Bengio. The smart buildings challenge: Learning foundation models for real-world optimization at scale. *arXiv preprint arXiv:2610.18511*, 2026.

[10] Judah A. Goldfeder and John A. Sipple. A lightweight calibrated simulation enabling efficient offline learning for optimal control of real buildings. In *Proceedings of the 10th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation (BuildSys '23)*, pages 352–356, 2023.

[11] Nikolaus Hansen. The cma evolution strategy: A tutorial. *arXiv preprint arXiv:1604.00772*, 2016.

[12] Thomas Krug, Fabian Raisch, Matthias Aimer, Michael Wirnsberger, Stephan Sigg, Maximilian Schäfer, and Benjamin Tischler. Builda: A thermal building data generation framework for transfer learning. *arXiv preprint arXiv:2508.12703*, 2025.

[13] Danil Malyshev. Smart buildings probe: zt-best. Kaggle notebook, NeurIPS 2026 Smart Buildings Challenge, 2026. `https://www.kaggle.com/code/malyshevdanil/smart-buildings-probe-zt-best`.

[14] Meteostat. Meteostat: Open access weather and climate data. `https://meteostat.net`, 2024. Licensed under CC BY 4.0.

[15] Zoltan Nagy, Gregor Henze, Sourav Dey, Javier Arroyo, Lieve Helsen, Xiangyu Zhang, Bingqing Chen, Kadir Amasyali, Kuldeep Kurte, Ahmed Zamzam, et al. Ten questions concerning reinforcement learning for building energy management. *Building and Environment*, 2023.

[16] NeurIPS 2026 Smart Buildings Challenge organizers. Smart buildings challenge: Rules and submission limits. Kaggle competition rules, `neurips-2026-smart-buildings`, 2026. Wall-clock budget of 9 hours for 30 scenarios, stated as ten times the bang-bang baseline's wall-clock; `https://www.kaggle.com/competitions/neurips-2026-smart-buildings/rules`.

[17] Kingsley Nweye, Zoltan Nagy, Sharada Mohanty, Dipam Chakraborty, Siva Sankaranarayanan, Tianzhen Hong, Sourav Dey, Gregor Henze, Jan Drgona, et al. The citylearn challenge 2022: Overview, results, and lessons learned. In *Proceedings of the NeurIPS 2022 Competition Track, PMLR 220*, pages 85–103, 2023.

[18] Fabian Raisch, Thomas Krug, Christoph Goebel, and Benjamin Tischler. Gentl: A general transfer learning model for building thermal dynamics. In *Proceedings of the 16th ACM International Conference on Future and Sustainable Energy Systems (e-Energy '25)*, 2025. arXiv:2501.13703.

[19] Tim Salimans, Jonathan Ho, Xi Chen, Szymon Sidor, and Ilya Sutskever. Evolution strategies as a scalable alternative to reinforcement learning. *arXiv preprint arXiv:1703.03864*, 2017.

[20] StnCsn. Starter kit PR #2: Energy-balanced hot-water tank. GitHub pull request, `smart-buildings-2026/starter-kit`, 2026. `https://github.com/smart-buildings-2026/starter-kit/pull/2`.

[21] StnCsn. Starter kit PR #3: Mix return air by airflow, not floor area. GitHub pull request, `smart-buildings-2026/starter-kit`, 2026. `https://github.com/smart-buildings-2026/starter-kit/pull/3`.

[22] Vincent Taboga, Justin Veilleux, Doseok Jang, Anushree Rankawat, and Pierre-Luc Bacon. Building2building: A large scale benchmark for generalizable real-world reinforcement learning. In *Reinforcement Learning Conference (RLC)*, 2026. arXiv:2607.16534.

[23] Steven T. Taylor. Resetting setpoints using trim & respond logic. *ASHRAE Journal*, 57(11): 52–57, 2015.

[24] Daniel Wu. Analytic MPC with a learned drift-residual ensemble for HVAC control. Kaggle notebook `danielwuu/cpu-a-alpha075-control-20261022`, NeurIPS 2026 Smart Buildings Challenge, 2026. `https://www.kaggle.com/code/danielwuu/cpu-a-alpha075-control-20261022`.

A The seven-day anchor

Table 2: The seven-day anchor the public forum compares on (`building_train_1/fairbanks_1`, 2 016 steps, mean outside $-26.3\,^{\circ}\mathrm{C}$, one seed), in reward units (higher is better). A positive Δ is better (reward units). Public rows are other teams' published numbers on the same episode, not re-run by us, and we assume the default occupancy seed; *foresight* is analysis-only. CtrlMPC is our receding-horizon LP on a 79-node twin, v1 as first written and v2 with its three defects fixed.

Controller	anchor total	Δ vs CtrlW	note
CtrlW (ours)	-93.12	—	readiness + weights + guards
CtrlW4 (ours)	-93.13	-0.004	pre-dawn lift, inert here
CtrlMPC v1 (ours)	-100.20	-7.08	8 h LP on the 79-node twin
CtrlMPC2 (ours)	-94.15	-1.02	LP defects fixed, no release found
CtrlW5-278 (shipped)	-86.68	$+6.44$	rooms $\geq 4.66\,^{\circ}\mathrm{C}$
CtrlW5-262 (unconstrained)	$\mathbf{-80.31}$	$\mathbf{+12.82}$	rooms to $-8.58\,^{\circ}\mathrm{C}$
perfect per-zone foresight	-90.52	$+2.60$	analysis only
physical public controller [20]	-81.9	$+11.2$	published
twin + LP public planner [1]	-85.4	$+7.7$	published
free-action LP bound [1]	-79.1	$+14.0$	published bound

B Ablations

Table 3: Ablations against the public base controller on dev30 (shipped), one change at a time, with the comfort component ($\times 10^{-4}$ per step) and the better/identical/worse count over the 30 scenarios (against the base for a–d, against CtrlW for the last four rows). h is the preheat horizon. CtrlW is a–d together and is 4.6×10^{-5} better than the horizon change alone, an interaction the one-at-a-time rows do not capture.

Variant	score	Δ	comfort	better / same / worse
base (Wu et al.)	0.0075040	—	2.14	—
(a) occupancy release	0.0075040	0	2.14	0 / 30 / 0
(b) reward-consistent fuel weights	0.0075035	-5×10^{-7}	2.13	16 / 10 / 4
(c) holiday guard	0.0075040	0	2.14	0 / 30 / 0
(d) readiness $h = 2$ h	0.0074263	-7.8×10^{-5}	0.55	23 / 0 / 7
(d) readiness $h = 3$ h	0.0074906	-1.3×10^{-5}	0.31	16 / 0 / 14
(d) readiness $h = 4$ h	0.0075614	$+5.7 \times 10^{-5}$	0.30	10 / 0 / 20
CtrlW (a–d, $h = 2$ h)	0.0073694	$-1.8\,\%$	0.55	30 / 0 / 0
CtrlW2 (per-zone horizon)	0.0073576	$-0.16\,\%$	0.59	12 / 12 / 6
CtrlW4 (+2 K pre-dawn lift)	0.0073445	$-0.34\,\%$	0.87	8 / 12 / 10
CtrlW5-278 (+release, 278 K)	**0.0071980**	$-2.3\,\%$	1.06	18 / 12 / 0
CtrlW5 (+release, 262 K)	0.0071338	$-3.2\,\%$	1.41	18 / 12 / 0

C The information ladder

Table 4: The information ladder behind §6. Scores are the mean over scenarios of $|\frac{1}{T}\sum_t r_t|$ (lower is better): *subset* is ten scenarios at full episode length (its baseline is 0.0119199 because it over-represents the coldest station, so only deltas compare across rows), dev30 the 30 shipped scenarios, *anchor* the seven-day comparison episode. Rows 3–7 read channels the competition does not expose and are analysis-only; row 2 needs no channel and is a legal controller. †the 1 K morning relax is a negative result, included because it measures the first-arrival asymmetry directly.

#	what the controller knows	subset	Δ	dev30	anchor
1	nothing (baseline)	0.0119199	—	0.0073694	-93.12
2	+ fixed 2 K pre-dawn lift	0.0118597	$-0.50\,\%$	0.0073463	not run
3	zonal occupancy, 5 min late	0.0116456	$-2.30\,\%$	0.0071585	not run
4	zonal occupancy, 289 K floor	0.0116352	$-2.39\,\%$	not run	not run
5†	zonal occupancy + 1 K morning relax	0.0118677	$-0.44\,\%$	not run	not run
6	perfect 12 h weather forecast	0.0118603	$-0.50\,\%$	0.0073572	not run
7	perfect per-zone foresight (5 h lead)	0.0112001	$-6.04\,\%$	not run	-90.52

D The accounting audit, in full

The measurements behind §7, on the seven-day anchor, one seed. Both quantities are valid only at the instant the reward is billed, so they are captured by wrapping the model calls rather than by re-reading sensors (every read of the boiler's supply-water sensor advances the tank), and a policy's own trajectory is not perturbed: three rollouts of one episode with full hooks, with only the reward wrapper and with nothing give bit-identical per-step rewards (-93.12290938248043 shipped, -92.78747206777916 corrected), and on the corrected tree the instrument reports $+0.00$ kW of return-air imbalance (largest residual 2.2×10^{-13} W), reproducing the published fix rather than approximating it. The *worth* column is shipped minus corrected, i.e. what the artifacts are worth to that trajectory (negative means the controller is *better* on the corrected accounting). *Free heat* is the B2 imbalance a policy realises, as a share of the coil heat it buys.

Table 5: Accounting audit on the seven-day anchor (`building_train_1/fairbanks_1`, 2 016 steps, one seed), shipped tree against the tree with both public fixes. The upper block is the controllers that score well; the probes are analysis-only.

Policy	shipped	corrected	worth	free heat	gas billed (MWh)
ours (CtrlW)	-93.123	-92.787	-0.34	182 kWh (0.7 %)	$27.19 \to 27.09$
Wu et al. (base)	-95.003	-94.581	-0.42	183 kWh (0.7 %)	$27.33 \to 27.21$
per-zone PI (zt-best)	-105.173	-104.317	-0.86	303 kWh (1.0 %)	$30.05 \to 29.64$
organizer `pid_water`	-596.865	-763.390	$+166.53$	1724 kWh (1.7 %)	$95.76 \to 124.53$
B1 probe (sawtooth)	-56.189	-93.380	$+37.19$	182 kWh (0.7 %)	$16.06 \to 26.98$
B2 probe (linear ramp)	-243.341	-255.838	$+12.50$	2370 kWh (4.6 %)	$51.38 \to 53.95$
B2 probe (2 kPa)	-4707.708	$-10\,258.068$	$+5550.36$	40 810 kWh (184 %)	$22.37 \to 32.82$

E Exactness and speed, in full

Table 6 lists what each acceleration buys and at what exactness. The measurements behind it: the fused step's twelve closed-loop gate cases plus full 8 641-step traces on both trees; the five Python patches (cached Woodbury response replacing a 19.5 ms triangular solve with a 2.7 ms matvec, memoised zone means, memoised schedule and timezone queries, hoisted per-zone reward metadata, a cached gather index) at 1.6–1.9× for +170 MB per worker, inert when the fused step owns the thermal solve; the solver study (the per-step system is symmetric positive definite with $\kappa \approx 1.4 \times 10^{-3}$ and a Jacobi contraction bound of 0.9989, so an unpreconditioned iteration needs about 2 500 iterations for 10^{-12}, while a direct factorization built once per episode costs 0.2–1.2 s); the batched sweep saturating at $M = 32$–64 because it becomes main-memory bound (42 GB/s measured), with $M = 16$ already capturing 90 % of the win and 19 of 20 gated trajectories identical (in the twentieth a damper decision flips at step 2 564 on a 10^{-14} field difference, and the episode total ends 3.5×10^{-9} relative away); the GPU verdict (smoothed-aggregation multigrid with a fixed 16-iteration PCG, 9.0×10^{-14} to 1.03×10^{-13} relative per step, 2.15×10^{-13} worst case chained, 22.3–23.5 ms per environment-step, the deficit toolchain rather than algorithm, a fused V-cycle projected at ≈ 1.2 ms still behind the CPU path, and SuperLU's own batched solve *degrading* from 7.3 to 8.7 ms per environment-step as M goes from 1 to 64); and the reduced model (projection in the energy inner product $u = C^{1/2}T$, a plain Galerkin projection of $A^{-1}C$ producing spurious eigenvalues outside the unit disc and leaving the physical range within twelve steps, 1.0–1.2 K maximum drift and a -0.2 to -0.3 K bias flat from $m = 100$ to $m = 400$ that refitting does not remove, closed-loop reward off by 2.6 % and 5.0 % while the five-way ranking is exact, thermal step 14× cheaper, end to end only 1.19×).

Table 6: What each acceleration buys, at what exactness. *Exactness* is measured against the unmodified simulator: a per-step relative error plus whether the episode total and the policy's decisions are unchanged. Speed-ups are interleaved A/B ratios on one machine; absolute times move with load, every ratio reproduced across sessions and on a second CPU.

Path	mechanism	speed	exactness	use
fused C step	one C call per step	2.1/1.8/1.6×	rewards/obs. bit-identical, 12/12	our runner
`-fast`	5 memoizations	1.6–1.9×	$\leq 2.8 \times 10^{-13}$ K; totals equal	local sweeps
LL^T, one RHS	METIS Cholesky	0.9×	3.7×10^{-14}	keep SuperLU
LL^T, batched	M right-hand sides	4.1–4.9× per env-step	3.7×10^{-14}/solve; 19/20 equal	RL rollouts
GPU MG-PCG	16-iteration PCG	0.3× of 1 core	1×10^{-13}/step, 2.2×10^{-13} chained	no-go
ROM, $m = 200$	energy-norm projection	14× solve, 1.19× e2e	1.0–1.2 K; ranking exact	shortlisting

F The bill, decomposed

The instrumented decomposition behind §3 (Figure 2), per full-length episode, in reward units (positive = loss). The outside-air column is the same step re-evaluated with the mixing box recirculating everything, i.e. the exact marginal effect of the outside-air fraction, with no assumption about coil saturation. Re-deriving cost and carbon from the logged powers reproduces the reward's own terms (4.9×10^{-10} relative worst case), and all six episodes reproduce the plain runner bit-for-bit.

Table 7: Energy-plus-carbon loss by mechanism, six climates (CtrlW, 8 641 steps each). Heating dominates the two Fairbanks buildings and chiller work the three warm ones; the outside-air share of the coil load is 50 % and 63 % in the two heating weeks, yet 100 % recirculation would cut coil duty by only 18.6 % and 9.3 %, because the coils are water-limited—the remainder is met by the fabric and appears as colder zones, one physical loss charged twice.

Scenario	$\bar{T}_{\text{out}}$	heating: OA	heating: envelope	boiler	cooling	fans	pump	total
Fairbanks, 39 zones	−21.4	69.65	306.69	5.08	0.00	8.62	2.51	392.55
Fairbanks, 85 zones	−8.2	9.29	90.55	2.12	0.00	4.06	0.66	106.69
Denver, 36 zones	24.2	0.00	0.16	1.56	32.70	2.97	0.00	37.39
Atlanta, 85 zones	20.8	0.01	0.39	0.91	19.44	3.83	0.00	24.58
Phoenix, 30 zones	34.4	0.00	0.00	0.07	78.06	4.87	0.00	83.00
Singapore, 54 zones	28.6	0.00	0.00	0.08	55.05	3.51	0.00	58.65

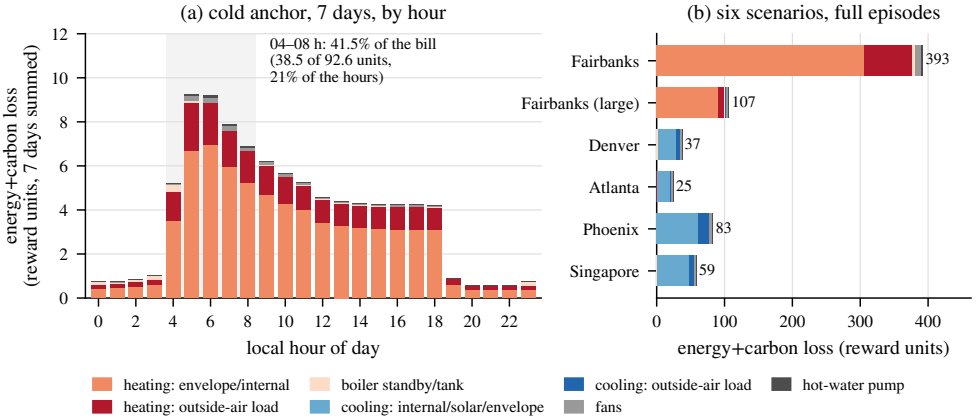


Figure 2: Reward anatomy for a controller whose comfort term is already 0.6 % of its loss. **(a)** The public anchor: energy-plus-carbon loss by local hour, stacked by mechanism—envelope reheat (light) plus the outside-air share of the same reheat (dark), 41.5 % of it in the capacity-limited 04–08 h block, 21 % of the hours. **(b)** The same decomposition for one full-length episode per climate; three scenarios hold 83 % of the 702.9-unit total (Appendix F). Each bar is a partition of the reward's own terms, reconciled below 1×10^{-9} relative.

The warm stations: nothing to release. The release is inert on the warm stations because both conditions it exploits are heating-only. On three seven-day warm anchors (CtrlW5-278, shipped) the chiller is 93–95 % of the bill and the plant is already off when the building is empty (Phoenix: 0.5 % of chiller energy at night, 0.0 % at weekends). The coil is billed at a constant COP of 3.5 with no capacity limit, so there is no analogue of Eq. (2): only airflow matters, and both airflow floors bind—the supply-air setpoint sits at the 285 K bound of the action space on 100, 97.5 and 91.8 % of live AHU-steps (Phoenix, Singapore, Atlanta) and recirculation at the 0.70 device cap on 97–100 %. Raising the unoccupied ceiling is bit-identical (it is never reached); riding the band's top 0.12 K costs 5.0 units on Phoenix, because the margin is the tracking error and the reward prices an occupied kelvin at 298 units a week against 0.62 of cooling; pre-cooling into cheap hours loses 3.9–6.9, because the electricity term is price-normalised (moving all of a week's cooling to its cheapest hour is worth 0.91 of 24.6 units); a night flush is inert where nights are warm and loses where it fires; and supply-air, reheat and static-pressure resets are outside the action space.

G Frost exposure

The measurements behind §4.5, on the six `dev30` Fairbanks scenarios at 8 641 steps, recorded by stepping the episode exactly as the evaluation runner does and reading only the zone-average temperatures (no monkeypatching, no extra reward call; the recorder reproduces the evaluation total to four decimals in every configuration where both exist). *h-any* is hours in which any room is below the threshold, *zone-h* room-hours.

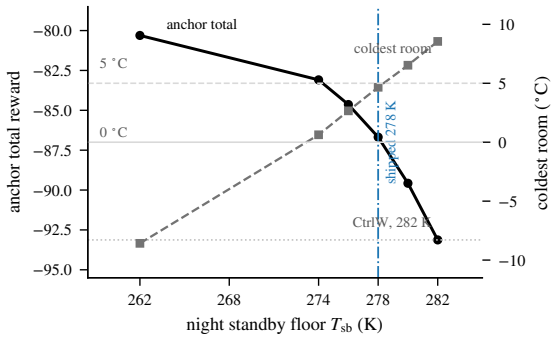


Figure 3: Price of a frost floor on the anchor (2 016 steps): anchor total reward (left) and the coldest room temperature the episode reaches (right) against the night standby floor, CtrlW5's other knobs fixed. Zero freezing is reached at 278 K; a $\geq 5\,^{\circ}$C setpoint floor needs 280 K.

Table 8: Frost exposure and its price on the three cold `dev30` Fairbanks scenarios (building_train_1,2,8/fairbanks_{1,2,8}, 8 641 steps each); the other three never leave the band, so the floor is inert there. *h-any* is hours in which any room is below the threshold, *zone-h* room-hours, and *mean* the mean of $|\frac{1}{T}\sum_t r_t|$ over the three. Over all six Fairbanks scenarios the set scores are 0.015653 (CtrlW), 0.015063 (CtrlW5-278, $-3.8\,\%$) and 0.014733 (CtrlW5, $-5.9\,\%$).

Controller	coldest room	h-any $< 5\,^{\circ}$C	h-any $< 0\,^{\circ}$C	zone-h $< 0\,^{\circ}$C	mean	Δ vs CtrlW
CtrlW (282 K)	$+8.53$	0.0	0.0	0	0.027436	—
CtrlW5-278 (shipped)	$+4.56$	459.8	0.0	0	0.026384	$-3.8\,\%$
CtrlW5-262 (uncontrolled)	-11.21	496.1	274.9	2681.8	0.025724	$-6.2\,\%$

H Reproducibility

Result directories contain one JSON per scenario with the score components, the hour-of-day and weekday/weekend decomposition and the metering-hook output; every entry in Tables 1–8 is read from them, and Figures 1–2 are drawn from stored rollouts by `paper/figures/{multiplier,frost,anatomy}.py`, with no simulator calls. The base controller is ported from the public notebook [24] under Apache-2.0; with our flags off it reproduces its author's dev30 score to the last digit (0.0075040), which is the parity check we ran before adding anything.

Provenance. The shipped tree is the starter kit at commit `95cea5b` (2026-10-19, including the tracker fix of PR #1); the corrected tree adds PR #2 and PR #3 [20, 21], frozen as patches and applied by `sbx/make_sims.sh`. Generated buildings are the first 24 kit-generator seeds from 5000 that fall inside the shipped zone and air-handler ranges (5001–5025 for `gen_train`, 5026–5046 for `gen_hold`), materials seed 20261105 (`sbx/make_pool.py`); the external strips are described in Appendix I. Occupancy runs on the kit's seed 17321, and the reseeds use 12345 and 777. The anchor CMA-ES (`cma` 4.5.0) searched CtrlW4's 16 knobs scaled to $[0, 1]$ at $\sigma_0 = 0.05$, population 12, seed 7, for 150 evaluations, starting from the best of 40 random points, with the default point injected in every population.

Commands. With `PY=starter-kit/.venv/bin/python` in the repository root ([pending: URL, with the code release]):

```
$PY -m sbx.evalrun --policy policies.ctrl_w5:CtrlW5 --params '{"standby_minimum": 278.0}' \
    --set dev30 --steps 8641 --sim shipped --tag ctw5frost_dev30_shipped
$PY -m sbx.evalrun ... --sim corrected            # the corrected tree
$PY -m sbx.evalrun ... --set gen_hold ext_hold    # the 49 holdouts
$PY -m sbx.evalrun ... --cenv fused --fast        # the exact C step, same totals
$PY -m cenv.verify.gate trace --scenario building_train_8/fairbanks_8 --cenv fused --out t8_c
$PY -m kaggle.remote.remote_eval --policy ... --set dev30 --shards 4 --cenv fused --fast
$PY kaggle/build_notebook.py --policy policies/ctrl_w5.py:CtrlW5 \
    --param standby_minimum=278.0 --slug sbx-ctrl-w5f   # the leaderboard entry
```

Machines and numerics. Local runs used an Apple M3 and a Ryzen 5 5600X (WSL2); the 30-scenario and holdout sweeps also ran on Kaggle CPU kernels, at most four at a time. The fused step is compiled with `-O3 -ffp-contract=off -fno-fast-math`—no FMA contraction, no reassociation—the flags under which it is bit-identical to the Python step on each machine. Across machines (arm64 with Accelerate against x86-64 with OpenBLAS) the same episode differs by up to 3.3×10^{-5} relative over 30 days—round-off amplified by the closed loop, e.g. -93.12290938 against -93.12336623 for the seven-day anchor—far below the printed precision of S and the reseed spread, but enough to turn "identical" into "better" or "worse"; every better/identical/worse count therefore compares runs from one machine.

I External weather

`ext` replays hourly observations from Meteostat [14] for calendar year 2023 at seventeen stations absent from the benchmark—Anchorage, Auckland, Bangkok, Beijing, Chicago, Dubai, Helsinki, Houston, Las Vegas, London, Miami, Montreal, Moscow, Reykjavik, San Francisco, Seattle and Tokyo—mapped to the benchmark's weather-file schema with no other scenario field modified. Meteostat data are distributed under CC BY 4.0; derived weather files are redistributed under the same terms with attribution.